

References:

- <https://math.stackexchange.com/questions/30046/alternative-notation-for-exponents-logs-and-roots/165225#165225>
- <https://youtu.be/sULa9Lc4pck?si=w3hDuLxaEmNoFqfE>
- <https://youtube.com/shorts/JNVZK4ofW-Y?si=zoxH2b5XLUfAKn4K>

Chapter 0: Binary Operations, Levels 1 and 2

Let's talk about arithmetic a little bit. There's this idea in mathematics of a binary operation. Here, "binary" refers to 2 things, and "operation" refers to doing something with them. Is that clear? Probably not. First of all, the "things" we are talking about are going to be numbers, at least at first. The stuff we are doing with them (the operations) will probably be familiar to you (addition, subtraction, multiplication, etc.). Each of those operations takes in two numbers and spits out a number. Some binary operations have some useful characteristics, and we're going to talk about those next.

Your identity is probably very important to you, right? It is who you are. Binary operations sometimes have something called an identity element. Here, "element" just refers to something you can use the binary operation on (usually a number). The identity element is the thing that just leaves everyone else exactly who they are. For example, addition has an identity element: zero. You can add zero to anything, and no changes occur. Anybody who gets added to zero gets to keep their own identity. Can you figure out the identity element in multiplication?

You ever see that episode of the original Star Trek where there are evil versions of characters, and you can tell they're evil because they have goatees? OK, well what about in Nintendo games, where there's sometimes an evil version of a character, and the evil one's name starts with a W? No? OK, never mind. Anyway, some binary operations come with this neat feature where each element, each number you can feed into it, has an inverse element. You can tell when one thing is an inverse of something else by combining them; when you use the binary operation to combine something with its inverse, the result is always the identity element.

So, remember how, with addition, the identity element is 0? What could you add to 5 in order to get 0? You could add -5 . For addition, the inverse of a number is always just the negative of that number. For multiplication, since the identity element is 1 (did you figure that out?), when you multiply something by its inverse, the result is 1. This one is a little trickier: in multiplication, the inverse is the reciprocal of the number. Are you familiar with that term? It means when you flip a fraction upside down. For example, 5 is the same as $\frac{5}{1}$, so the reciprocal of 5 is $\frac{1}{5}$. Those two numbers are multiplicative inverses of each other, so $5 \times \frac{1}{5} = 1$. Here's another example: $\frac{-2}{3} \times \frac{-3}{2} = 1$.

I work from home, so I don't drive to work anymore. This is a good thing, because I am a terrible driver. Anyway, people who travel to and from work every day, especially if they travel a long way, are called commuters, and "to commute" basically means to go back and forth. In mathematics, some binary operations also commute, though it means something slightly different. If a binary operation is commutative, that means you can reverse the order of the two numbers you are operating on (a.k.a. the two input elements), and the result doesn't change. Addition is commutative (notice that $5 + 4$ gives the same result as $4 + 5$). Multiplication is also commutative. You know what's not commutative, though? Subtraction. Changing the order of the numbers in subtraction changes the result. For example, $9 - 3$ is not the same as $3 - 9$. The result from the first one is 6, and from the second one is -6 .

That's kind of interesting, though. In subtraction, when you swap the order of the elements, the output changes to the additive inverse of what it was originally. For example, $1.06 - 9.5$ gives the additive inverse of $9.5 - 1.06$ (Did the decimals throw you off? Don't worry, they're still just numbers). This is not commutative, but it feels kinda similar. Let's call it co-inversative.

There's another weird thing about subtraction. Although you can't swap the two inputs and leave the output the same, you can swap the right input element with the output. By "right", I mean right versus left. For example, consider the equation $100 - 30 = 70$. The right input is 30, and it can be swapped with the output to make a new equation: $100 - 70 = 30$. However, if you try to swap the output with the left input, it doesn't work:

70 – 30 is not 100. Since you can only swap the right element with the output, I am going to say that subtraction is right co-inversative.

So, addition and subtraction go together, you know? I like to think of them as Level 1 Operations. There is a third Level 1 Operation, and you probably know how to do it, even though it doesn't get taught in school. I call it Fromation. How do you do it? It's super easy. Take 2 from 6, and what's left? Did you say 4? Then you know how to from.

Here's another example: 13 *from* 29 = 16. Now, try taking the three numbers in that fromation equation and swapping them around to see if it still works. Seriously, do it.

Did you do it? If so, you might have noticed that if you swap the two inputs, the output does not stay the same; it becomes the additive inverse of the original output, so fromation must be co-inversative. You also might have noticed that if you swap the output with the left input, the new statement is still true: 16 *from* 29 = 13. Therefore, fromation is left co-inversative.

Even through the paper on which this is printed, I can hear you screaming that fromation is just subtraction. Go soak your head. Two *from* five is three, but two *minus* five is negative three.

OK, now that you've got that out of your system, let's take a look at a table summarizing the Level 1 Operations.

	Addition	Subtraction	Fromation
Forwards	$2 + 3 = 5$	$5 - 2 = 3$	2 from 5 = 3
Backwards	$3 + 2 = 5$	$5 - 3 = 2$	3 from 5 = 2
	Commutative	Right Co-inversative	Left Co-inversative

Let's move on to Level 2 Operations. We already talked about multiplication, which is commutative. What about division? Is that commutative? Nope. Here, take a look: $12 \div 3 = 4$, but $3 \div 12 = \frac{1}{4}$. If you swap the inputs, the output changes. Definitely not commutative. On the other hand, check this out: 4 and $\frac{1}{4}$ are multiplicative inverses! If you multiply them, you get 1, which is the identity element for multiplication. Test out a few more examples, and you should be able to convince yourself that division is co-inversative. So, is it left co-inversative or right co-inversative? Try a few examples and see if you can figure it out. Use a calculator if you want. Did you figure it out? It's right co-inversative.

However, we should note that there is an exception to this kind of thing at Level 2. At this level, zero has the special property of just destroying everything it touches. This does not affect multiplication's commutativity, but it does affect division. The right input element in division cannot be zero (i.e. no dividing by zero), at least not as long as we are dealing with the traditional number system. So yes, it *is* right co-inversative, but with that one exception.

If you are the kind of person who notices patterns, you might guess that there is another Level 2 Operation. You're right, there is: Gozintation. I bet you already know how to do it. For example, can you tell me how many times 2 gozinta 14? Did you say 7? If not, try reading it aloud: "Two gozinta fourteen". OK, so let's explore it. Write out a gozinta equation. You can use 2 *gozinta* 14 = 7, or you can come up with your own. Do it. Seriously, write it down. Now try rearranging the numbers. Can you swap the inputs? No? Well, then it's not commutative. If you do some careful exploration, however, you might discover that it is left co-inversative. For example, check this out: 3 gozinta 6 two times, and 6 gozinta 3 half of a time. Two and one-half are multiplicative inverses. Also, 2 gozinta 6 three times; we can swap the left element with the output. Therefore, gozintation is left co-inversative.

I'm just going to pretend I don't hear you yelling about how gozintation is the same as division. Go ahead, take all the time you need. When you've calmed down, come on back, and let's take a look at a table summarizing the Level 2 Operations.

	Multiplication	Division	Gozintation
Forwards	$2 \times 3 = 6$	$6 \div 2 = 3$	2 gozinta 6 = 3
Backwards	$3 \times 2 = 6$	$6 \div 3 = 2$	3 gozinta 6 = 2
	Commutative	Right Co-inversative	Left Co-inversative

Something that will come up later is the idea that these binary operations are really more like trinary relationships. At level 1, you have 2, 3, and 5 that all go together. At level 2, the numbers 2, 3, and 6 form the same

kind of trio. However, before we investigate that further, we need to step away from just numbers and bring in some letters.

Chapter 1: Unwrapping Mathematical Gifts

Imagine you have just received a present. There's a ribbon and wrapping paper and maybe even a card. Beneath the wrapping paper, there's probably a box. Inside the box, there might be some tissue paper surrounding the present. If the person giving you the gift thinks they are really funny, there might be another fully wrapped box in there. So, how do you go about finding out what the present is? You unwrap it, right? Don't just shake the box; what if it's a puppy!? Let's talk about unwrapping.

Often, in algebra, one is asked to find the value of an unknown quantity using some information about that quantity. Here's an example:

There is a number, x , such that three times x is equal to twelve.
Determine the value of x .

Now, you might be able to just, like, *know* the answer to that one. However, I don't care about that. What I am interested in here is describing the **process** whereby one might determine the value of x if it were not obvious from the beginning. The way this is often taught (and which I assume you have seen before) is by writing an equation and then performing changes upon that equation, which changes are restricted in that they must have the same effect on both sides of the equation. Oof, that's a lot of words. Instead, I shall give you some more words: "I'm going to ... both sides ...".

The two ellipses in that sentence get replaced with whatever you want to do in order to unwrap your present. Let's take a look at the setup I gave you before. Since three times x is equal to twelve, we can write that in mathematical notation.

$$3x = 12$$

Now, the first thing to do here is to ask yourself whether x has been isolated yet. Has it been totally unwrapped, or are there still some numbers around it? In this case, the x is not yet alone. There is a 3 next to it. Cool. Now, we want to get rid of that 3, to peel it off of the x so we can see what's inside. But how do we do it? Well, check out how the 3 is attached to the x . It is attached by multiplication. In order to get rid of multiplication, we can do the opposite: division. Therefore, I'm going to divide both sides by 3.

$$\begin{array}{rcl} 3x & = & 12 \\ \div 3 & & \div 3 \\ \hline x & = & 4 \end{array}$$

Now that x is completely unwrapped, we can see that it is equal to 4. Cool. Why does that work? Well, we did the same thing to both sides of the equation, so the effect was the same, so if the two sides were equal before, then the two new sides must be equal afterwards. Let's take a look at another example.

$$x - 5 = 19$$

How do we open this present? Well, is the x alone? No. What else is there? There's a 5. How is it attached to the x ? Via subtraction. How do we cancel out subtraction? With addition. Therefore, I am going to add 5 to both sides.

$$\begin{array}{rcl} x - 5 & = & 19 \\ +5 & +5 & \\ \hline x & = & 24 \end{array}$$

So, let's think about this another way. Instead of performing an action to both sides of an equation, what if we think about it as two equations, and we are adding them together, right side to right side and left side to left side? Let's see what this looks like.

$$\begin{array}{rcl} x - 5 & = & 19 \\ + & & 5 = 5 \\ \hline x & = & 24 \end{array}$$

Now, why would I do that? Well, the idea here is to show that “I’m going to” statements are just simple, convenient ways of discussing what’s really happening. Deep down, what we are doing is combining two equations. In this case, we combined them using addition. Consider the fact that both of the starting equations are true (the second one is obvious, and the first one is based on given information). If we are just combining two true equations, then it kind of doesn’t matter which one we start with. Check this out:

$$\begin{array}{r} 5 = 5 \\ + \quad x - 5 = 19 \\ \hline x = 24 \end{array}$$

OK, but so what, right? Well, the rule here, the general principle, is that, if you have two true equations, then you can combine them (left to left and right to right) to make another true equation. In this case, this happens to be useful if we are trying to solve an equation by unwrapping x . The weird thing about this second way of writing it is that the “I’m going to” statement changes a little. Here, we might say something like I’m going to add both sides to 5.

Next, let’s take a look at a situation where we start with an addition problem involving x .

$$x + 1,000 = 2,345$$

If we were doing an “I’m going to” approach, you could say that you were going to subtract 1,000 from both sides. That would certainly work. You could also think of writing the equation $1,000 = 1,000$ (which is obviously true), and then finding a way to combine it with the equation you already have. In this case, you would probably use subtraction.

$$\begin{array}{r} x + 1,000 = 2,345 \\ - \quad 1,000 = 1,000 \\ \hline x = 1,345 \end{array}$$

Just in case you want to see that in even more detail, check this out. I’ll do the thing I did before where I highlight the left side in one color and the right side in another color, but this time I am going to be more explicit about what happens when the left is combined with the left and the right with the right.

$$\begin{array}{r} x + 1,000 = 2,345 \\ - \quad 1,000 = 1,000 \\ \hline (x + 1,000) - (1,000) = (2,345) - (1,000) \\ x = 1,345 \end{array}$$

Now, we’ve looked at a couple of examples, and I hope you found those doable. However, there are some more complicated ones that come up, even when there’s only one binary operation involved, and even when it’s a Level 1 Binary Operation. Take a look at this example:

$$27 - x = 2$$

Now, we want to get x by itself, but x is being subtracted from 27. A common way to look at this is to say that x is negative, and then we can add x to both sides and subtract 2 from both sides and then end up with x all by itself on the right. However, I would like to offer other options. First, consider the fact that subtraction is right co-inversative. Wait, do you remember what right co-inversative is? That was covered in the last section. It looks like this:

$$\begin{array}{l} S - M = D \text{ implies that } S - D = M \\ \text{for example:} \\ 30 - 6 = 24 \text{ implies that } 30 - 24 = 6 \end{array}$$

Because subtraction is right co-inversative, can swap the right element with the output and arrive at a new, true statement.

$$27 - x = 2$$

SWAP!

$$27 - 2 = x$$

Not bad, right? Here's another way to look at it. Do you have any neighbors? That might be a loaded question, because you might live in the same building with your neighbors, or there might be no one else living on the entire rural road where you make your home, or you might be reading this in a post-apocalyptic hellscape where the very concept of "neighbors" is confusing. In order to avoid those concerns, do me a favor and pretend you live in a house, that the house is on a very long, completely straight road, that there are other houses on that same road, and that people live in those other houses. Now, pick one of these imaginary neighbors and consider how far it is from your house to your neighbor's house.

Wait, why were we doing that?

OK, so the road is like the number line, right? And maybe your house is at #27, and your neighbor is somewhere else, at a place called x . That's a strange name for a house, I'll grant you, but this entire situation is imaginary, so it's fine. Anyway, the distance from your house to your neighbor's house might be expressed as " $27 - x$ ".

OK, enough metaphors. Now just think about " $27 - x$ ", all by itself, just as a mathematical expression.

Wait, do you know what a mathematical expression is? It's like an equation, but without the equals sign. It's a bunch of numbers and variables and whatnot connected by operations, but you can't "solve" it because you don't know what it's equal to. OK, let's get back into it.

Think about " $27 - x$ " as a mathematical expression. That's the difference between 27 and x , right? Like, x is in one place, and 27 is somewhere else, and this is kind of like how far apart they are. Ignore for a moment that you know how far apart they are (or did you forget? Remember we are talking about the equation $27 - x = 2$), and just think of it as " $27 - x$ ". Now, if we were to subtract that distance, that " $27 - x$ " (whatever it might be) from 27, we would move that distance from 27. And since that distance is how far you have to go from 27 to get to x , by subtracting it from 27, we would get to x . Hooo boy, that sounded way better in my head. Going back to the imaginary neighbors idea, if you started at your house and walked along the road for a distance equal to how far Bob's house is from yours, then (as long as you were walking in the correct direction) you would end up at Bob's house. That really didn't help, did it?

Let me put it another way. I suggest that 27 is equal to itself. We can agree on that, right? And also, we can agree that we can combine two true equations using subtraction and arrive at a new true equation, right? Well, take a look at this:

$$\begin{array}{r} 27 = 27 \\ - \quad 27 - x = 2 \\ \hline x = 25 \end{array}$$

OK, so what happened there? Well, if we think about starting with 27 and then we subtract " $27 - x$ ", we end up with a positive x . Is that strange? Well, remember that subtracting something is the same as adding the negative version (the additive inverse). So, we can think of subtracting " $27 - x$ " as adding the negative of " $27 - x$ ". We would write that as " $-(27 - x)$ ". The negative sign is outside of the parentheses because we are applying it to the whole expression. To get rid of the parentheses, we just have to apply that negative sign to each of the terms inside. That way, " $-(27 - x)$ " turns into " $-27 - -x$ ", which is the same as " $-27 + x$ ".

$$\begin{array}{r} 27 = 27 \\ - \quad 27 - x = 2 \\ \hline (27) - (27 - x) = (27) - (2) \\ 27 - 27 + x = 27 - 2 \\ x = 25 \end{array}$$

In an "I'm going to" paradigm, what we did here was take both sides **from** 27. More specifically, you could say I'm going to take both sides from 27. As such, we could have written it like this:

$$\begin{array}{r} 27 - x = 2 \\ \text{from } 27 \quad = 27 \\ \hline x = 25 \end{array}$$

Now, realistically, nobody is going to write it that way. However, what I'd like to suggest is that you can think about subtracting both sides from the same value. With addition, the order of the two equations didn't matter; whether we added 5 to both sides or we added both sides to 5, the result was the same. However, because subtraction is not commutative, swapping the order of the elements made the output turn into the additive inverse of what it originally was. If we were to subtract 27 from "27 - x", we would end up with -x, but if we subtract "27 - x" from 27 (that is, if we reverse the order), we get the opposite, we get positive x.

OK, we've covered what happens when something is being multiplied by x, and when something is being added to x, and when something is being subtracted from x, and when x is being subtracted from something. Since addition and multiplication are commutative, we only had to cover one approach for each of them, but for subtraction, we had to consider whether x was the right or left element. We're going to move on to division now, and since division is not commutative, there will again be two different situations to consider. Let's start with the easier one.

$$\frac{x}{7} = 31$$

By the way, when expressing division, I will occasionally switch back and forth between using the $\div$ symbol (also known as the **obelus**, which is a word also used for a lot of other symbols) and writing it as a fraction, where the dividend is the numerator and divisor is the denominator. Man, knowing all the terminology really just doesn't help much, does it? Anyway, returning to that equation, what should we do? We could say I'm going to multiply both sides by 7. We could also introduce the equation $7 = 7$ (which is obviously true) and write it either above or below the other equation, and then multiply the left sides by each other and the right sides by each other.

$$\begin{array}{r} \frac{x}{7} = 31 \\ \times \quad 7 = 7 \\ \hline x = 217 \end{array}$$

$$\begin{array}{r} 7 = 7 \\ \times \quad \frac{x}{7} = 31 \\ \hline x = 217 \end{array}$$

That really isn't so bad, right? I hope? However, it gets a little more complicated when the present we are trying to unwrap is hidden under something else. In this case, I mean when the variable occurs as the second element in the division (that is, when it is the divisor or the denominator). Let's take a look at one of those:

$$\frac{15}{x} = 3$$

OK, so, in school, you have probably been taught a few different ways to approach this kind of thing. Many of you have probably learned to take the 3 and turn it into $\frac{3}{1}$ so you get a proportion (an equation where each side is a fraction), and then "cross multiply" or "do the propellor thing". I hate those. You wind up having to do another step afterward. There's only one binary operation here, so we should be able to unwrap this present with just one step. Let's use the other way of writing division and look at it again.

$$15 \div x = 3$$

So, we know that division is right co-inversative, just like subtraction, so we can swap the right input with the output.

$$\begin{array}{l} 15 \div x = 3 \\ \text{SWAP} \\ 15 \div 3 = x \end{array}$$

That's pretty cool, but again, I want to think about combining equations. What is the thing we want to get rid of here, the wrapping paper that is covering up the present? It's 15, right? So we should be able to use the equation $15 = 15$ (which is obviously true), combined with the equation involving x, to get x by itself. We can do

that, but just like when x was being subtracted from something, we have to go about it in a weird way. Let's write $15 = 15$ above the equation involving x , and then use division to combine them.

$$\begin{array}{r} 15 = 15 \\ \div \quad 15 \div x = 3 \\ \hline x = 5 \end{array}$$

Again, this is probably an unfamiliar approach to most of you. Here's a way to think about what happened on the left side.

$$\begin{array}{l} 15 \div (15 \div x) \\ 15 \div \left(\frac{15}{x}\right) \\ 15 \left(\frac{x}{15}\right) \end{array}$$

What we did in the last step there was to remember that dividing by something is the same as multiplying by its reciprocal, its multiplicative inverse. Therefore, we can switch from division to multiplication if we turn the right element upside-down. Anyway, going back to combining the two equations using division, in the "I'm going to" paradigm, we would say that I'm going to divide 15 by both sides.

(I am just going to pause here and say that it is taking all of my self-restraint to avoid inserting an entire section about how the ways in which you can rewrite a proportion are the same as the symmetries of a square. I mean, it's super cool, but it's not what we're talking about here.)

You know, I think maybe all of that just wasn't satisfactory. We don't just want to know that we *can* divide 15 by both sides; we want to know *why* we would do that. Let's take a look at the equation again.

$$15 \div x = 3$$

Now, go with me on this one, OK? When we ask "what is fifteen divided by x ?", we could also ask "how many times does x go into 15?" and we'd be looking for the same answer. $15 \div x$ is the same as x *gozinta* 15. Wait, I know! Let's do a numerical example, and I'll put boxes around the parts I want to talk about.

$$\boxed{28 \div 4} = 7 \qquad \boxed{4 \text{ gozinta } 28} = 7$$

Right, so, think about the thing in the box. The box represents a number. I mean, it equals 7, but the way I want you to think about it is "how many times four gozinta twenty-eight". So, if I were to take "28" and divide it by the "number of times $\boxed{4 \text{ gozinta } 28}$ ", what would I get? What would you get? Hopefully, we would get the same thing. Twenty-eight divided by the number of times four gozinta twenty-eight should be four, right? Okay, let's write that out as math.

$$28 \div \boxed{4 \text{ gozinta } 28} = 4$$

Cool. Now let's jump back to the equation from before this one. I'm just going to write the left-hand side, and I'm also going to write it as a gozinta.

$$\boxed{15 \div x} \qquad \boxed{x \text{ gozinta } 15}$$

Now, think about that thing in the box. If I were to take 15 and divide it by $\boxed{x \text{ gozinta } 15}$, what would I (and also you) get?

$$15 \div \boxed{x \text{ gozinta } 15} = x$$

Good, good. OK, now let's look at the whole equation.

$$15 \div x = 3$$

Rewrite it as a gozinta.

$$x \text{ gozinta } 15 = 3$$

Now let's divide 15 by both sides of that equation and then simplify.

$$\begin{aligned} 15 \div (x \text{ gozinta } 15) &= 15 \div (3) \\ x &= 5 \end{aligned}$$

Side note for mathematicians: fromation and gozintation are their own inverse operations.

OK, I hope that worked for you, because now it's time to take this to Level 3. Look at the following equation:

$$x^2 = 25$$

This one is telling you that there is some number (x) which, when you square it (or "raise it to the second"), gives 25 as the output. This operation, "raising" or "powers" or "exponents" or "to the", is the first of the Level 3 Operations. I'm guessing that you were taught at some point that the opposite of squaring is called square rooting. Let's look at that for a sec, and say I'm going to take the square root of both sides.

$$\begin{aligned} x^2 &= 25 \\ \sqrt{x^2} &= \sqrt{25} \\ x &= \pm 5 \end{aligned}$$

If you are thrown off by the $\pm$, just remember that every positive number has two square roots, one positive and one negative. Cool. Maybe you've even been taught that the opposite of raising to the n is taking the n^{th} root.

Now, let's consider what happens if we start with x inside a radical.

$$\sqrt[3]{x} = 3$$

Have you seen the root with the tiny little number? It means the 3rd root (or "cube root"), and it's the inverse of raising to the 3rd (cubing). Now, I'm not usually going to say "square root" and "cube root". Instead, I am going to say the number of the root, by which I mean the tiny little number sitting on the shelf on the left side. For example, when you see $\sqrt[5]{99}$, you can read that as "the fifth root of ninety-nine". Also, when it's a "square root" or "second root", people usually leave out the 2, but you can put it in there if you want.

Anyway, getting back to our 3rd root equation, let's raise both sides to the 3 and see what happens.

$$\begin{aligned} \sqrt[3]{x} &= 3 \\ (\sqrt[3]{x})^3 &= (3)^3 \\ x &= 27 \end{aligned}$$

That was OK, but it seems to have deviated from what we were doing before, right? Like, why do the elements sometimes have to be written in different sizes when we are dealing with Level 3 Operations? The lack of consistency irks me. Also, the fact that we write it differently disguises the fact that other ways of writing equations with Level 3 Operations are very hard to solve. For example:

$$2^x = 128$$

Again, maybe if you are really good with your powers of 2, you can figure out what x is supposed to be there, but also again, what I am trying to do is talk about either unwrapping a present (isolating x by doing the

same thing to both sides of an equation) or combining two different equations, left to left and right to right. What “I’m going to” statement could we use here? What other equation could we combine with this one, and what operation would we use to combine them? We are now entering a troubling area, where the presents are not only wrapped up, but stored in a puzzle box with no obvious way to get it open. To deal with this new level of difficulty (Level 3), we are going to have to introduce a fancy new tool: the TRIANGLE OF POWER.

Chapter 2: Introducing the Triangle of Power

2A: The way things are right now kind of stinks

The standard notation for Level 3 Binary Operations is pretty bad. Really, the only operation at that level that has something that even comes close to a normal operation symbol is raising. Oh, did I mention that the words available to describe the operations are also terrible? For comparison, let's review the Level 1 and Level 2 Operations. For each one, I will give the name of the operation (e.g. Addition) and also the word(s) we say when we read it (e.g. plus).

Addition	Subtraction	Fromation
$2 + 3 = 5$ plus	$5 - 2 = 3$ Minus	$2 \text{ from } 5 = 3$ from
$3 + 2 = 5$ plus	$5 - 3 = 2$ Minus	$3 \text{ from } 5 = 2$ from
Also Addition (Commutative)	Also Subtraction (Right Co-inversative)	Also Fromation (Left Co-inversative)

Multiplication	Division	Gozintation
$2 \times 3 = 6$ times	$6 \div 2 = 3$ divided by	$2 \text{ gozinta } 6 = 3$ goes into
$3 \times 2 = 6$ times	$6 \div 3 = 2$ divided by	$3 \text{ gozinta } 6 = 2$ goes into
Also Multiplication (Commutative)	Also Division (Right Co-inversative)	Also Gozintation (Left Co-inversative)

If you look carefully, you might notice that I've set up those tables to follow a particular pattern. In the top left cell, I start with 2 as the left input, 3 as the right input, and use the simplest operation at that level to create an output. The cell below it has the inputs swapped (which works because multiplication and addition are both commutative).

In the second column, the thing that was the output in the first column (5 or 6) is placed first, as the left input. For the second row, the output is swapped with the right input (which works because subtraction and division are both right co-inversative).

In the third column, the thing that was the output in the first column is placed second, as the right input. In the second row, the left input is swapped with the output (which works because fromation and gozintation are both left co-inversative).

Did I really need to say all of that? Or should I have said "I mean, just look at it!"?

Altogether, each table shows all possible ways of arranging three numbers, left to right, with an operation between the first two and an equals sign between the last two. It's all the ways you can write something of this form:

NUMBER OPERATION NUMBER EQUALS NUMBER

Let's try to build the Level 3 table and see what happens (spoiler: this is not going to go well). For the very first cell, the one in the top left, let's use the fact that $2^3 = 8$. Now, you might already notice that there's no operation symbol here, but one does exist. You can often see it on the keypad of a calculator. We don't usually write it out, but if we did, it would look like this: $2 \wedge 3 = 8$. OK, let's start the table. I refer to this operation as Raising, and claim that, when we read it out loud, we say "to the".

Raising		
$2 \wedge 3 = 8$ to the		

--	--	--

So far, our table looks very empty. Next, I am going to fill in the numbers 2, 3, and 8 in their various orders, following the pattern we used for the Level 1 and Level 2 tables.

Raising		
$2^3 = 8$ to the	$8 \div 2 = 3$	$2 \div 8 = 3$
$3 \div 2 = 8$	$8 \div 3 = 2$	$3 \div 8 = 2$

At this point, you might be tempted to go through and replace those question marks with symbols, but what symbols would you use? Also, what headings would you put on each column, like what name would you give to each binary operation? Also also, notice that none of these will be commutative or co-inversative, so we would actually need *six different operation names*. That's just straight up nutty. Schools already don't bother to teach the names of one of the three operations at each of the first two levels; it's unlikely they'll want to teach *six* different operation names for Level 3. Unless you've done some advanced math, the only one of those you might have a symbol and name for is the one in the bottom right.

Raising		
$2^3 = 8$ to the	$8 \div 2 = 3$	$2 \div 8 = 3$
$3 \div 2 = 8$	$8 \div 3 = 2$	$\sqrt[3]{8} = 2$ root of

But what do we call that operation? Rooting? Radicalation? Extraction of surds? More like *absurd*. None of these are terribly satisfying. Personally, I like to call it lowering, but I might be the only person who does, and I don't even do it consistently (you will probably see me call it rooting). Now, just so you know, I'm not really going to give names to the operations. It would just be a whole bunch more vocabulary, and it wouldn't really help you understand what's happening, and also no one would ever agree with the terminology I pick. Instead, what I am going to do is assign names to the **role** each number plays in the relationship, in the trio. In the equation $2^3 = 8$, the 2 will be called the base. 3 will be called the exponent. 8 will be called the power.

$$\begin{aligned} \text{base}^{\text{exponent}} &= \text{power} \\ \text{base}^{\text{exponent}} &= \text{power} \end{aligned}$$

Take a moment to remember where the 2, 3, and 8 go in the root (lowering) form. That's the one in the lower right in the (mostly empty) table of Level 3 Operations. We can rewrite this as follows.

$$\text{exponent} \sqrt{\text{power}} = \text{base}$$

If you are yelling at the paper that those aren't the vocab terms you are used to, hey man, just chill. Roll with it. I'll bring up those other words later. For right now, though, I do want to mention one thing. The most frequent confusion people make is using the word "power" when they mean "exponent". This is all related to the fact that mathematics has failed to provide us with good, consistent vocabulary in this particular area. Do you remember when computers used to come as a bunch of different pieces? Like, there was a monitor and a mouse and a keyboard, and those were all "peripherals", and then there was a box that had the motherboard and the hard drive and the floppy disk drive and the CPU, and that box was the "computer". If you haven't seen a computer like that, just trust me. Anyway, because the monitor was where you would spend all of your time looking, people used to refer to the monitor as "the computer". This is not correct, but people just tended to run with it. Also, because people were calling that part the computer, they needed a different word for the box that actually held the computer, so they started calling that "the CPU". This is also not technically correct, but usually it didn't create any

problems because people could figure out what someone meant from the context. However, if someone said, without context, “I need a new CPU”, it wasn’t entirely clear whether they actually need a new CPU or a whole new computer (but not a new monitor or keyboard). It is unlikely that anyone in the target audience will relate to this example.

OK, so, connecting this story back to the math stuff, the whole expression “ 2^3 ” is a power. One might say “eight is the third power of two”. However, because we express powers using a base and an exponent, and the exponent is kind of the weird part, people will sometimes call it a power. Don’t call it a power. No one will know whether you are referring to the monitor or the CPU or the computer or the whole shebang.

2B: The way things could be is much better

OK, now that we have names for the jobs each of those numbers is doing in the equation, we’re going to take a look at a really cool way of writing these operations: the Triangle of Power. Let’s start with that very first equation/relationship, $2^3 = 8$.

$$2^3 = 8 \text{ means } 2^3 = 8 \text{ means } \begin{array}{c} 3 \\ \triangle \\ 2 \end{array} = 8$$

In general, if we want to express raising, like when we have something *to the* something else, we put the **base** at the lower left corner of the triangle and the **exponent** at the top.

$$\begin{array}{c} \text{exponent} \\ \triangle \\ \text{base} \end{array} = \text{Power}$$

One way to think about this is that, if you remove the Triangle of Power, it just looks like $2^3 = 8$. Cool. Since the Triangle of Power here is equal to a Power, I am going to refer to things like 2^3 as power expressions. On the other hand, for stuff that looks like $\sqrt[3]{8}$, I will use the term root expression. Now, if we want to write a root expression using the Triangle of Power, we actually use the exact same symbol, but just change where we put the pieces.

$$\sqrt[3]{8} = 2 \text{ means } \begin{array}{c} 3 \\ \triangle_8 \\ \end{array} = 2$$

In general, if we want to express lowering (or rooting or whatever you want to call it), we put the **power** at the lower right corner of the triangle and the **exponent** at the top.

$$\begin{array}{c} \text{exponent} \\ \triangle \\ \text{power} \end{array} = \text{Base}$$

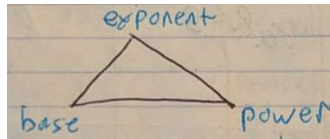
In case you would like this in symbols (instead of words or a numerical example), here you go:

$$\begin{array}{c} y \\ \triangle \\ x \end{array} \text{ means } x^y$$

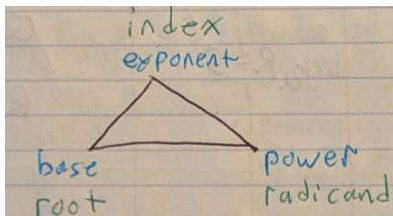
$$\begin{array}{c} x \\ \triangle_y \\ \end{array} \text{ means } \sqrt[x]{y}$$

So, two things. First of all, writing out “Triangle of Power” is exhausting, so henceforth I am going to use the abbreviation ToP. You can pronounce this like “top” if you want to, but in my head, I say “tea oh pea”. Second, the really neat thing about the ToP is that, if you use all three corners, it acts as an entire equation. Not only that, it acts as two different equations at the same time. Remember the relationship between 2, 3, and 8? We can express that by filling in all three corners. When all three corners of the ToP are filled in, it represents an equation.

When a ToP represents an equation, the places where you put the base, the exponent, and the power do not change; you just fill in all three.



For those of you who are not comfortable using those words where there's a root expression involved, thank you for your patience. Please remember that the terms themselves are not what is important. The important thing is what role they play in the relationship. Therefore, if you really want to, you can rename the corners when you think of it as involving a root.



Now, I know that thinking about this symbol as an equation is weird. Usually, when you have an equation, you have an equals sign, right? If you really want to, you can draw an equals sign inside the ToP, but I don't think it'll help. Anyway, when you see all three corners of the ToP filled in, you can choose to think of it as either the root equation (root expression = base) or the power equation (power expression = power). Honestly, it doesn't really matter, because they are saying the same thing. It's kind of like when you say Dave is the son of Erlizabeta, and Erlizabeta is the mother of Dave. It doesn't matter whether you are talking about who is whose mother or who is whose son; the relationship is the same. Eight is the third power of two, and two is the third root of eight.

Oh hey, remember earlier, back at the end of chapter 0, when I said it made more sense to think about trios of numbers instead of binary operations? This right here, the thing about how 8 is the 3rd power of 2, and 2 is the 3rd root of 8, this is where I was going with that idea.

2C: Why is this section about grammar?

So, hey, did you learn grammar in school? I think most people don't anymore, largely because of terrible misinterpretations of the theories of Noam Chomsky, but that story is way outside the scope of what I'm trying to do here. Anyway, in sentence structure, when you have a verb, which is a word for an action (usually), it can often have both a subject and an object. The subject is the thing/place/person/idea that is *doing* the action, and the object is the thing/place/person/idea upon which the action is performed. For example, in the sentence "Tigger ate Piglet", the subject is Tigger and the object is Piglet. Sorry, was that a little dark? Here's a different example: in the sentence "The Argonauts followed Jason", the subject is the Argonauts and the object is Jason. An interesting thing about that one is that we can use Jason as the subject if we pick a different verb. In "Jason led the Argonauts", Jason has become the subject.

In mathematics, there's a related concept of making something the subject of an equation. Generally, this involves solving for the thing, i.e. unwrapping it. If you isolate a variable on one side of an equation, you have made it the subject. This kind of description is particularly useful when the equation has more than one variable. Let's look at an example.

$$Q = R + V$$

Right now, Q is the subject because it is all by itself. However, we can use “I’m going to” statements to switch things around. What steps would you have to go through to make R the subject? Can you do it in one step?

$$R = Q - V$$

I find that people feel more comfortable when the subject of an equation is written on the left side. I wonder if that’s related to the fact that I mostly interact with people who speak English, and sentences in English tend to have the subject on the left, before the verb and the object? That might be going a bit too far into the grammar analogy, though, so let’s move on.

So, when we rearrange an equation to change the subject, let’s call that resubjecting. One of the oddities of the ToP is that, when all corners are filled in so that it is acting like an equation, the subject is a little ambiguous. Like, is the subject the *power*, or is it the *base*? The best way to think about this, I think, is that the ToP expresses the relationship between the base, the exponent, and the power without specifying the subject. If you convert a ToP into a more traditional form, you have to choose a subject. Something really useful happens here; if you start with a traditional equation, convert it into a ToP, and then convert it back to a traditional equation, you don’t have to keep the subject the same. You can resubject the equation just by moving in and out of ToP form.

The diagram shows the following steps: $5^4 = 625 \rightarrow 5 \triangle^4 = 625 \rightarrow 5 \triangle^4_{625} \rightarrow \triangle^4_{625} = 5 \rightarrow \sqrt[4]{625} = 5$. The $\triangle$ symbol represents the ToP (Top Power) notation.

Let’s talk about what just happened there. We started with an equation with a traditional power expression. We replaced the 5^4 with its ToP form. Then we moved the 625 into the ToP, so the ToP became an entire equation. Then, we pulled the 5 out and made it the subject. Finally, we converted the ToP notation into its more traditional form (when the exponent and power are both present), as a root expression. That sounds like a lot of steps, but once you become comfortable with it, you can do it in just two.

The diagram shows the following steps: $5^4 = 625 \rightarrow 5 \triangle^4_{625} \rightarrow \sqrt[4]{625} = 5$.

Again, we started out with 625 as the subject... Wait. Didn’t I just say a little while ago that people tend to be more comfortable when the subject is on the left side? OK, let me rewrite this. Hold please.

The diagram shows the following steps: $625 = 5^4 \rightarrow 5 \triangle^4_{625} \rightarrow 5 = \sqrt[4]{625}$.

OK, so, we started out with 625 as the subject, and did two things. First, we transferred the equation into ToP form. Second, we transferred back into traditional form, but making 5 the subject. What did we accomplish? We resubjected the equation, but we didn’t actually have to perform any “I’m going to” steps at all. Maybe you think that’s cool, or maybe you don’t because you just have no sense of beauty. And maybe that last sentence was me drawing too extreme of a dichotomy.

2D: There is nothing rhythmic about these logs.

Something you might’ve noticed about me is that I like to explore all the different variations of things. That’s why I’m such a fan of fromation and gozintation; without them, there’s something missing. Well, there’s something missing here, too. We have seen how to take a ToP equation and resubject it to make either the base or the power the subject, but what about the exponent? Can you do that? Sure. However, this leads to another Level 3 Binary Operation, one you might not have seen before. It is called the logarithm. That is a terrible name, but even worse is the traditional notation for it. Here’s how it’s normally written:

$$\log_2 8 = 3$$

Isn't that hideous? I mean, why is the 2 relegated to the cellar? And if the operation is "log", why isn't it *between* the two input elements, the 2 and the 8? Ugh. It's awful. On the other hand, to write it in ToP form, we can just go ahead and use the exact same symbol again! Behold:

Handwritten ToP equations: $2 \triangle 8 = 3$ means $\log_2 8 = 3$

Now, that was a ToP equation, right? But just like with the other thingies we looked at, we can also use the ToP to represent a log expression.

Handwritten ToP equation: $x \triangle y$ means $\log_x y$

Some of you might be unsatisfied by the fact that there exists a word for this operation, and a way to write it. Maybe you'd like to know how one would go about calculating it. Well, like some roots and powers, it can be quite difficult to calculate, but we don't really need to know how to obtain the exact value in order to understand what it means. Just like children might use a multiplication table to figure out 12×7 , or any reasonable person would use a calculator to figure out $\sqrt[4]{13}$, we can make use of technology to compute logarithms.

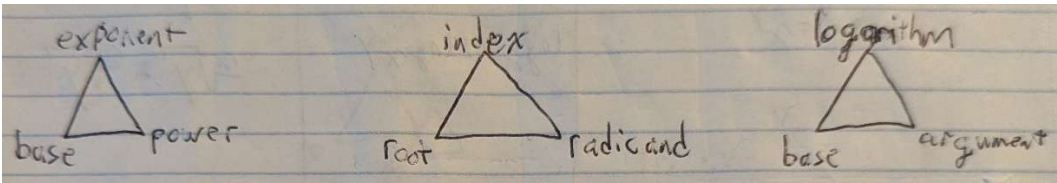
So, now that we have a name for this operation and a symbolic way of writing it, we can shove it into the table of Level 3 Operations.

Raising		Logarithm
$2^3 = 8$ to the	$8 ? 2 = 3$	$\log_2 8 = 3$ base log of
$3 ? 2 = 8$	$8 ? 3 = 2$	$\sqrt[3]{8} = 2$ root of
		Lowering

Please note that the words for the operation, like how you read it, are not exactly correct there. Technically, one does not say "Two base log of 8 equals 3". Instead, you might say "The base 2 log of 8 is 3" or "The logarithm, base 2, of 8 is 3". As long as we're talking about words, I might as well mention that the elements in a logarithm operation traditionally have their own names, unique to logs, even though they are still doing the same thing as they do in raising and lowering. Here are the names of the elements in the traditional way of writing a logarithm:

$$\log_{\text{base}}(\text{argument}) = \text{logarithm}$$

Yup, the output of a log operation is called the "logarithm". That is super confusing, right? Like, when you multiply two numbers, you don't call the output the "multiplication". Traditional log terminology is just gross. In case you want to know how the terms all correspond to each other, here's a diagram:



2E: Let's throw out the table and get back to unwrapping presents.

OK, so now we have three of the cells in the Level 3 Binary Operations table filled in. We are not going to do the rest of them because *they don't have names or symbols!* Isn't that crazy? Like, why, right? The lack of completeness just makes me feel itchy. However, instead of trying to invent terms and symbols, I propose that we

just trash all of that stuff and fill out the table like this, with a power expression, a root expression, and a logarithm expression:

Actually, that's a little silly. Let's just have one symbol for the entire Level 3 table.

Now, before we move on, there are three things about writing a ToP that we need to discuss: nesting, collapsing, and recentering. These are all related to the fact that, sometimes, the things in a ToP (i.e., the base, exponent, and power) are not just numbers, and they're not even variables. You see, sometimes a thing is actually two things, or three things, or even more. In my head, that last sentence sounded deep, but upon reading it, it just seems silly. Let's go into the math.

So, remember that whole thing about unwrapping presents? I mentioned that, sometimes, you have to unwrap several layers in order to actually get to the gift? Here's a math example of that:

$$2x^3 - 7 = 243$$

Here, if you want to get to the gift, there are several layers to remove. Just like with unwrapping a present, you should start with the outermost layer. That is, if x is not yet alone on its side of the equation, we should remove the other things also hanging around on that side, starting with the one "farthest" from x . The word "farthest" is not really a technical term here; I'm just using it as a metaphor because it works pretty well in this kind of situation. Let me explain.

So, the x is not alone, right? There are three other things wrapped around it: a 2, a 3, and a 7. The 243 doesn't count because it's not on the same side of the equation as x . In order to unwrap x , to tear off its mask and reveal its true self like a Scooby-Doo villain, we need to get rid of everything else that's on the same side of the equation, and that includes the 2, the 3, and the 7. OK, but which should we start with? The way I like to think about it is that you have to remove the outermost layer first, the thing farthest from the x . Now, the 3 is quite firmly affixed to the x , right? So that's the innermost layer. The next is the 2, which is right up against the side of the x . Finally, there's the 7, which is separated quite a bit from the x . OK, so we're going to get rid of the 7 first, and then the 2, and then the 3, and we're going to do each step with an "I'm going to" statement.

$$2x^3 - 7 = 243$$

I'm going to add 7 to both sides.

$$\begin{array}{r} 2x^3 - 7 = 243 \\ +7 \quad +7 \\ \hline 2x^3 = 250 \end{array}$$

I'm going to divide both sides by 2.

$$\begin{array}{r} 2x^3 = 250 \\ \hline \frac{2x^3}{2} = \frac{250}{2} \\ x^3 = 125 \end{array}$$

I'm going to take the 3rd root of both sides.

$$\begin{array}{r} \sqrt[3]{x^3} = \sqrt[3]{125} \\ x = 5 \end{array}$$

And here's my present: it's a 5. Yippee!

Anyway, when there are parentheses involved, sometimes there's a situation where there's a Level 3 Operation that has to be done as the *first step* because it's the outermost layer of wrapping paper. Take a look at this example:

$$(2x - 9)^3 = 343$$

First, I'm going to take the 3rd root of both sides.

$$\begin{aligned}\sqrt[3]{(2x - 9)^3} &= \sqrt[3]{343} \\ 2x - 9 &= 7\end{aligned}$$

I'm going to add 9 to both sides.

$$\begin{aligned}2x - 9 &= 7 \\ +9 &+9 \\ 2x &= 16\end{aligned}$$

I'm going to divide both sides by 2.

$$\begin{aligned}\frac{2x}{2} &= \frac{16}{2} \\ x &= 8\end{aligned}$$

I'd like to focus on the 3rd root step in the two solutions, and their similarities and differences. In both cases, we had something to the 3. That is, we had a power expression with a base and an exponent, and in both cases, the exponent was 3. It's just that the base looked different.

$$x^3 \quad \text{versus} \quad (2x - 9)^3$$

Now, even though the bases look different from each other, they still really have the same format:

$$\text{something}^3$$

That something could really be anything, right? It could be two things, or three things, or a whole lot more things. The outer layer would still be that exponent, the 3. Here's a silly example:

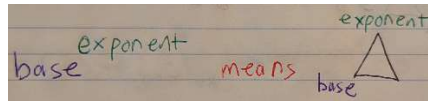
$$(a + 2b^2 + 3c^3 + 4d^4 + 5e^5 + 6f^6 + 7g^7 + 8h^8 + 9i^9)^3$$

If you were trying to solve for one of the variables, and you had an equation where one side consisted of that long power expression, the first thing you would probably do is take the 3rd root.

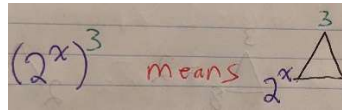
OK, so, we have seen that the base can be a whole expression, but that's also just as true of the exponent or the power. The rest of this chapter deals with situations where one or more of those three elements is an expression that also involves Level 3 Operations. Let's look at an example.

$$(2^x)^3 = 5$$

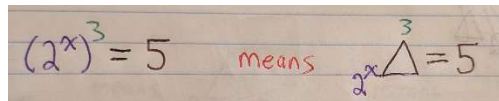
This time, we're not going to try to solve it right away. First, we're going to try to convert it to ToP form. Let's start with the left side. What we have there is a power expression, where the base (2^x) is being raised to an exponent (3). What does the ToP for that look like? First, remember how we can rewrite a power expression as a ToP.



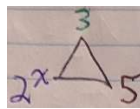
Next, substitute in the base and exponent.



Applying this to the equation means we can rewrite it like this:



Then, we can do the final step to include the other side of the equation as the missing corner of the ToP.

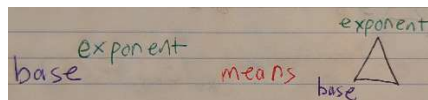


2F: ToPs on ToPs on ToPs

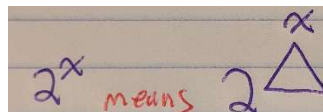
So far, what we have done here is basically the same thing as what we were doing before, except that the base isn't just a single number or letter. That thing is two things. The interesting thing about that thing being two things (this sentence is really something, huh?) is that we can make it into its own ToP. Let's zoom in on just that corner.



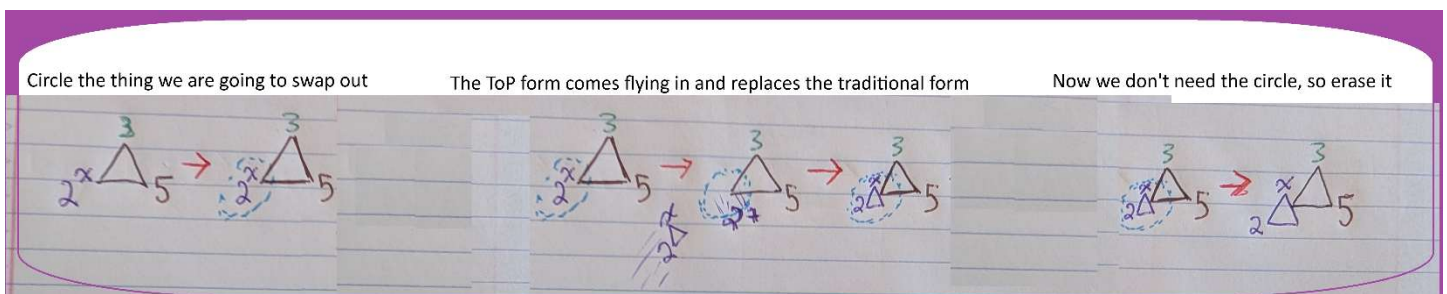
How would we rewrite that in ToP form? Well, let's go back to how we handle this kind of thing in general.



Applying that to 2^x , we get this:



Now, let's go back and look at the ToP equation we had, and replace the base with its own ToP.



If we want to be succinct about it, we can write it like this:

What we have just done here is called nesting. We took a ToP equation where one corner was a whole expression, and we turned that expression into its own ToP. It looks a little weird, I know, but once you get used to ToP notation, it will start to feel better. The opposite of nesting is called collapsing, and it just refers to taking the nested ToP and converting it back into traditional form. Since we can convert a simple ToP into three different forms of traditional equation, let's make 2^x the subject.

You know, that final form is kind of interesting, right? Like, what if we change the way we are looking at it. Here, I'll change the colors to show you what I mean. My apologies to those of you with color blindness, or actual blindness, or number-color synesthesia, or who just don't much care for color coding.

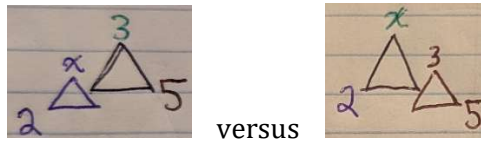
Now, another way to look at this same equation is that the base is 2, the exponent is x , and the power is $\sqrt[3]{5}$. Let's turn that into ToP form.

If you are getting good with it, maybe you can do that in one step.

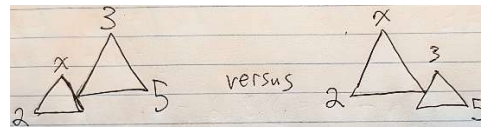
That root expression in the lower right corner, the power corner, do you see that? Do you see the $\sqrt[3]{5}$? Well, that can be put into ToP notation also, right?

OK, so let's use that to make a nested ToP.

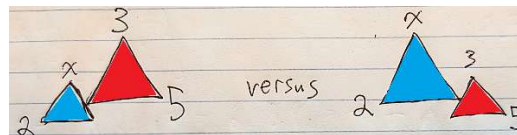
So, the weird thing about this is that it does not look like the nested ToP we got the first time. Let's compare.



Try to ignore the different color schemes. Actually, let me help with that.



OK, let's think about this for a minute. To help, I am going to add back in some color coding, but this time to help differentiate the two ToPs. Again, my apologies to those left out of the experience due to perceptual differences. The ToP with the 3 and 5 will be red. The one with 2 and x will be blue.



On the left side, thinking about the red ToP, the power is 5, the exponent is 3, and the base is the blue ToP, which is 2^x . But now, staying on the left side, look at the blue ToP. Clearly, the base is 2 and the exponent is x , but what is the power? OK, switch over to the right side and look at the blue triangle there. In this case, we can definitely see that the base is 2, the exponent is x , and the power is the red ToP, which is $\sqrt[3]{5}$. Huh. So, when a ToP appears nested inside another, it plays the role of wherever it's located, like how, on the right, the blue ToP is the base. On the other hand, the ToP within which it is nested plays the role of the missing corner, like how, on the right side, the red ToP is the power of the blue ToP. These things are interchangeable.

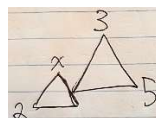
Man, that was a whole lot of words there, and we're not even done. This leads us to a new concept, requiring a new word: recentering.

2G: It all depends on how you look at it.

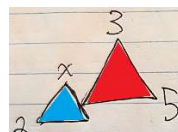
We already know that, in a ToP equation, you can make any corner the subject of the equation. In addition, when there is a nested ToP equation, you can choose any ToP to be the center of the whole thing. Switching the way you are looking at it – changing which ToP you think of as the center – is called recentering. When you combine it with resubjecting, some really fascinating stuff becomes possible. Let's go back to the equation that started our exploration of nested ToPs.

$$(2^x)^3 = 5$$

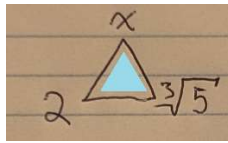
Through some manipulation, we got it into a nested ToP form, where it looked like this:



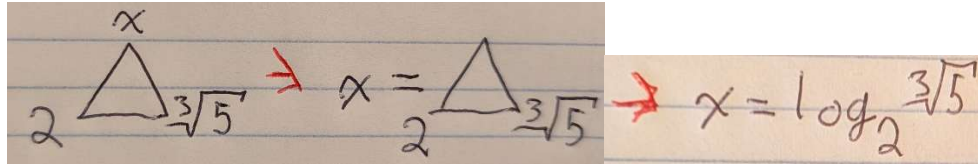
Let's put the color-coded ToPs back in (blue on the left, red on the right).



Now, the way we built this, the red ToP was the main one and the blue ToP was nested inside. Now let's recenter on the blue ToP. OK, done. No, seriously, you don't have to change anything except your perspective. Just look at the blue ToP as the main one. OK, right now, the power corner is a whole ToP, so let's collapse that.



Now, let's convert it into traditional form, but this time, let's make x the subject.



Let's take a step back and review what we just did. We took the equation $(2^x)^3 = 5$, converted it into a nested ToP, recentered, collapsed, resubjected, and converted back into a traditionally formatted equation to discover that $x = \log_2 \sqrt[3]{5}$. We solved for x without ever using a single "I'm going to" statement, and without ever combining our equation with another one. Seriously, you guys, this is like magic.

Chapter 3: Building Inverse Operations

3A: Raising and Lowering as inverses

Remember before how we were able to convert a ToP into three different looking equations, depending on which corner we chose to be the “subject” or “answer”? We called that resubjecting. Well, if we make some nested ToPs and resubject them in just the right way, we can arrive at some fascinating discoveries. Let’s start with an obvious power expression and convert it into nested ToP form.

$$B^x = B^x$$

In the obvious expression above, think of the right side of the equation as the subject, so, on the right side, B^x is the power, and on the left side, B is the base and x is the exponent.

$$B^x = B^x \rightarrow B^{\triangle x} = B^{\triangle x} \rightarrow B^{\triangle x} \triangle B$$

Obvious	Triangle
$B^x = B^x$	$B^{\triangle x} \triangle B$

We are about to take a little bit of a walk here. I acknowledge that this section might need a little cleanup. Please bear with me, and hopefully it will be clear within another page or two.

Now, in the nested ToP form, let’s stay centered on the left ToP. I’ll try to make it stand out a little more.

$$B^{\triangle x} \triangle B$$

The equation it came from, $B^x = B^x$, had the power corner as the subject. What if we make x the subject? Or B ? Remember, we are using the top left ToP. B is the base of that one, right? If we make B the subject, then we use the exponent and the power to form a root expression. Recall how that works:

$$\triangle_8^3 = 2 \text{ means } \sqrt[3]{8} = 2, \text{ and in general, } \triangle_P^x \text{ means } \sqrt[x]{P}$$

The only complication here is that the power is an entire ToP. It happens to represent B^x , right?

$$B^{\triangle x} \triangle B \text{ means } B^x$$

So the thing that goes inside the radical will be B^x .

$$\begin{array}{c} x \\ \triangle \\ B \end{array} \text{ means } \sqrt[x]{B^x}$$

OK, let's go back to the original obvious equation, convert it to a nested ToP, and then resubject around B .

$$B^x = B^x \rightarrow B^x = B^{\triangle x} \rightarrow B^x = B^{\triangle \triangle x} \rightarrow B = B^{\triangle x} \rightarrow B = B^{\triangle x} \rightarrow B = \sqrt[x]{B^x}$$

Now, this might not seem super exciting. However, consider that we started with an obviously true equation and did not actually change any values; we just moved stuff around and picked a different subject. That means the resulting equation is just as true as the original one. One of the useful things in math is that, if you know two things are equal to each other, you can substitute one for the other. Now we know that $B = \sqrt[x]{B^x}$, and we know that it's true no matter what B and x are. Therefore, if we ever find (or create) a situation where an equation has $\sqrt[x]{B^x}$ in it, or anything else of that form, we can replace it with just B . For example, look at this equation:

$$y^7 = 128$$

Yes, I know it doesn't have something that looks like $\sqrt[x]{B^x}$, but it will soon. Let's solve this using "I'm going to" statements. The thing we want to get rid of is the 7. I'm going to take the 7th root of both sides. Look what happens.

$$\sqrt[7]{y^7} = \sqrt[7]{128}$$

That got a whole lot uglier, but now check this out: the thing on the left side? We can replace it with just y . And the thing on the right side? You can just plug that into a calculator.

$$y = 2$$

Now, maybe you are saying that you already knew how to solve that kind of equation. That's wonderful! And also not the point! What I have done here is to demonstrate that rearranging an obviously true equation by making a ToP and resubjecting it can give us insight into solving equations. Our first insight is that if we have an equation with a power expression and we want to isolate the base, we can take a root of both sides.

What if we are given some other kind of expression with a Level 3 Operation? Let's look at root expressions first.

$$\text{exponent} \sqrt{\text{power}}$$

Root expressions have an exponent and a power. We'd like to know how to isolate either one of them, but let's start off trying to isolate the power. In order to figure out how to do that, let's start with an obviously true equation.

$$\sqrt[x]{P} = \sqrt[x]{P}$$

No matter what x and P are, the left and right sides of that equation must be equal, so that equation must be true. Let's convert it to ToP form, make P the subject, and convert it back to traditional form. First, here's the processes of turning it into a nested ToP.

$$\sqrt[x]{P} = \sqrt[x]{P} \rightarrow \triangle_P^x = \triangle_P^x \rightarrow \triangle_P^{\triangle x} = \triangle_P^{\triangle x}$$

Now, let's resubject around the P in the ToP on the right side, and then convert it back into traditional form.

Here, we have another interesting fact. We have learned that $P = (\sqrt[x]{P})^x$. This looks almost the same as the first one of these we got, but now the order of the lowering and the raising are reversed. The lowering is closer, a more interior layer of wrapping paper. The important thing here, though, is that we have learned that if we see $(\sqrt[x]{P})^x$, we can replace it with just P . When would that come up? Whenever we start with a root expression and want to isolate the base. Here's an example:

$$7 = \sqrt[3]{k}$$

We want to isolate the k , so we need to get rid of the 3. Since it's a root expression, I'm going to raise both sides to the 3rd.

$$\begin{aligned} 7 &= \sqrt[3]{k} \\ 7^3 &= (\sqrt[3]{k})^3 \end{aligned}$$

We can use our insight about powers and roots being inverses to replace the right side with k , and just multiply out the left side.

$$\begin{aligned} 7^3 &= (\sqrt[3]{k})^3 \\ 343 &= k \end{aligned}$$

OK, let's summarize what we just figured out.

1. $B = \sqrt[x]{B^x}$, so we can replace $\sqrt[x]{B^x}$ with B .
2. Therefore, if we have an equation where one side looks like B^x and we want to get B by itself, we can take the x^{th} root of both sides.
3. $P = (\sqrt[x]{P})^x$, so we can replace $(\sqrt[x]{P})^x$ with P .
4. Therefore, if we have an equation where one side looks like $\sqrt[x]{P}$ and we want to get P by itself, we can raise both sides to the x .

Again, I am aware that that was quite a trip, with a lot of symbol manipulation and whatnot. Here's the important thing: Do you believe items 2 and 4 in the list above? Do they make sense to you? And do you see how the stuff we did in this section shows why those methods work? If it's not clicking for you yet, maybe reread this section. Also, maybe I'll put together some exercises so you can play with it yourself. Or maybe I won't; I'm not a very industrious person.

3B: Isolating the exponent in a power expression

In the previous section, we started by writing out an obvious power expression: $B^x = B^x$. We then put it in ToP form and resubjected around B , and this led to an insight about how to solve equations with power expressions. However, it only told us how to solve for the base, not the exponent. Let's see if we can get a similar insight if we use that same equation as before, but resubject around x . I know that x is in there twice; I'm going to make the one on the left the subject.

$$B^x = B^x \rightarrow B^{\triangle x} = B^{\triangle x} \rightarrow B^{\triangle x} \rightarrow x = \log_B B^x \rightarrow x = \log_B B^x$$

That last one is that weird thing we learned in the previous chapter. Do you remember the traditional name for it? That operation is called logarithm. Here's a quick refresher:

$$\begin{matrix} 3 \\ \triangle \\ 2 \end{matrix} 8 \text{ means } 2^{\triangle 8} = 3 \text{ means } \log_2 8 = 3$$

OK, looking back at the thing we were working on, the right side has the format of a logarithm. Let's write it in the traditional format.

$$x = \log_B B^x$$

Gross. Anyway, this piece of information will come in handy in solving equations because we now know that, if we encounter (or create) $\log_B B^x$, we can just place that with x . Consider this equation:

$$16^z = 1,024$$

If you are the target audience for this thing you are reading, then you probably didn't know how to solve that equation before. However, the fact we just discovered gives us a potential approach. I'm going to take the base 16 logarithm of both sides.

$$\log_{16} 16^z = \log_{16} 1,024$$

The lefthand side now fits the format of the thing we just discovered, so it can be replaced with just z . Meanwhile, for the righthand side, you can just type that into a calculator (as long as you have a good one).

$$z = \frac{5}{2}$$

Of course, with something like that equation, it might be easier to just move it through ToP form to resubject it.

$$16^z = 1024 \rightarrow {}_{16}\triangle^z = 1024 \rightarrow {}_{16}\triangle^z_{1024} \rightarrow z = \log_{16} 1024$$

Here, we have a new insight. We already knew that, if we have something like B^x and we want to get B by itself, we can remove the exponent by taking the x root of it. Now, we also know that, if we want to get x by itself, we can remove the base by taking the base B logarithm of it. OK, so we know how to isolate either the base or the exponent when we are given a power expression. To summarize:

1. $x = \log_B B^x$, so we can replace $\log_B B^x$ with x .
2. Therefore, if we have an equation where one side looks like B^x and we want to get x by itself, we can take the base B logarithm of both sides.

3C: Advanced inverses

In the previous section, we learned how to isolate the exponent when we were given a power expression. That's fine, but how would we isolate the exponent in a root equation? Let's use that same obviously true root equation from before.

Now let's take that nested ToP and resubject around the exponent, x . There are two of them, so I'll circle which x I am talking about.

I mean, I guess that's a new true equation, but it's probably not clear how we would use that, right? Well, returning to the idea of substitution, if we ever happen to come across (or create) $x^x \Delta_P$, we would be able to replace it with just x . I will admit that this is one of the rarer circumstances, but it's still definitely possible. It might come up if you have a root expression and want to isolate the exponent. For example, let's say we want to solve the following equation for w .

$$\sqrt[w]{78,125} = 25$$

Guessing and checking would be a terrible approach to unwrapping w . Instead, we want to try to get rid of the 78,125 and leave w by itself. How do we do that? Well, remember that log is a binary operation, just like addition and subtraction and division, right? Now, if we were to say I'm going to take the base w log of both sides, that wouldn't really help. Saying that I'm going to take the base 78,125 log of both sides is a little better, but not much. However, there's a variation that will work. I'm going to take the base both sides log of 78,125. If that sounds strange, remember back in chapter 1 when we learned that with $27 - x = 2$, we could subtract both sides from 27 to get x alone? This is kind of like that. Anyway, the idea is that each side of the equation becomes the left element in the log operation, and the right element is 78,125 for both sides.

$$\begin{aligned} \sqrt[w]{78,125} &= 25 \\ \log_{\sqrt[w]{78,125}} \Delta_{78,125} &= 25 \Delta_{78,125} \end{aligned}$$

I'll also write it in traditional form. Isn't it awful?

$$\log_{\sqrt[w]{78,125}} 78,125 = \log_{25} 78,125$$

We can use that substitution rule we just discovered to turn the left side into w . As for the right side, we can just type that into a calculator.

$$w = 3.5$$

That one was kind of crazy, right? Well, we're not done yet. We've seen how, given a power expression, we can isolate the exponent or the base. We've seen how, given a root expression, we can isolate the exponent or the power. Finally, we're going to look at what happens when we are given a log expression and we want to isolate either the base or the power. As usual, we will start with an obviously true equation. I'll write it in traditional form first, but we'll switch to ToPs soon.

$$\log_B P = \log_B P$$

As with the previous two cases, let's first put it into nested ToP form.

$$\log_B P = \log_B P \rightarrow {}_B \Delta_P = {}_B \Delta_P \rightarrow {}_B \Delta_P = {}_B \Delta_P$$

OK, now let's resubject. To start, let's make P the subject. I'll circle the P that I am talking about.

$${}_B \Delta_P = {}_B \Delta_P \rightarrow P = {}_B \Delta_P \rightarrow P = {}^{\log_B P} B \rightarrow P = B^{\log_B P}$$

What insight have we gained from this? Well, now we know that, whenever we see something that looks like $B^{\log_B P}$, we can replace it with just P . Now, in what situation would that arise? Consider the following equation:

$$\log_2 x = 9 \quad \text{or} \quad {}_2 \Delta_x = 9$$

We want to get the x by itself, right? Hmm. If the whole of ${}_2 \Delta_x$ were somehow in an exponent instead of just sitting there, we could work with that. OK, so I want the left side to become $2^{{}_2 \Delta_x}$ (or $2^{\log_2 x}$ in old-timey notation), because then I could just replace it with x . How do I do that? What "I'm going to" statement could we use here? Take a moment to think about it.

If you didn't get it yet, consider the binary operation $^$. Do you remember that one? It means "to the", so if I write 5^3 , that means the same thing as 5^3 ; if I have something like z^q , I can rewrite that as z^q . OK, now try to use that idea somehow to come up with an "I'm going to" statement. Take a moment to think about it.

If you didn't get it yet, let's try this. We want to turn the left side into $2^{{}_2 \Delta_x}$, right? But as discussed in the last hint, we could rewrite that as $2^{\log_2 x}$. Do you see the binary operation there? Which part was already in the equation, and which part do we have to use the "I'm going to" statement to bring in?

If you didn't get it yet, I'm all out of hints, so I'm just going to tell you. I'm going to raise 2 to both sides. Let's walk through what that means. First of all, can we agree that 2 is equal to itself? I hope so. If not, maybe this should be a different discussion. Anyway, assuming that we agree that 2 is equal to itself, we can write the following equation:

$$2 = 2$$

Great. I'm glad you're coming with me on this. Now, the whole idea behind the "I'm going to" statements is that, if you do the same thing to both sides of a true equation, then the new equation you create is also true, right? So, for example, I could raise both sides of that equation to the 3rd.

$$2^3 = 2^3$$

Good, good. Now, when we do something like this, where we apply an exponent to both sides of the equation, the exponents we use don't have to look exactly the same, as long as they are equal. Here's an example:

$$2^{4.5} = 2^{\frac{9}{2}}$$

Since $4.5 = \frac{9}{2}$, what we did there was take the true equation ($2 = 2$), and raise both sides to equal exponents. Everybody still with me? OK, cool. Now, remember the logarithm equation we were working on a minute ago? Here it is again:

$${}_2 \Delta_x = 9$$

Now let's do that "I'm going to" statement to it. I'm going to raise 2 to both sides.

$$2 = 2$$

$$2^{\Delta x} = 9$$

$$2^{\Delta x} = 2^9$$

Now we can simplify the left side using the insight we had about things that look like that, and we can simplify the right side using a calculator (or pencil and paper if you are feeling brave).

$$\begin{aligned} 2^{\Delta x} &= 2^9 \\ x &= 512 \end{aligned}$$

Nice. Now we are almost done. I mean, not with everything, but with this particular thing. Let's take a look back at the nested ToP we are working with right now. Do you remember this one?

Earlier, we went through the process of making P the subject. What if we try to make B the subject? I'll circle which B I mean.

That is a crazy looking thing, but it provides a new insight. If we ever come across (or create) something that looks like $B^{\Delta P} \sqrt{P}$, we can replace it with just B . We'd want to do that if we had a logarithm equation and were trying to solve for the base. Here's an example:

$$y^{\Delta_{216}} = 3$$

Is the y alone? No it is not. What is there with it? 216. This means that we probably want to use the fact that $216 = 216$. How would we do it? What would we do to both sides? Take a moment to think about it.

Did you think about it? No? Seriously, take a moment. Go back and review the previous example; it might help you come up with ideas.

Well, whether you thought about it or not, here's the solution. I'm going to take the both sides root of 216. Again, this means that the two sides of the original equation are going to be the exponents in the new equation. The difference here is that they will be on root expressions rather than power expressions.

$$216 = 216$$

$$y^{\Delta_{216}} = 3$$

$$y^{\Delta_{216}} \sqrt[216]{216} = \sqrt[216]{216}$$

Nice, nice, looking good. Now we can use our recent discovery to simplify the left side. On the right, we can proceed with a calculator (or with pencil and paper if you know how to do cube roots by hand).

$$\begin{aligned} y^{\Delta_{216}} \sqrt[216]{216} &= \sqrt[216]{216} \\ y &= 6 \end{aligned}$$

OK, let's summarize what we just figured out.

1. $x^{\Delta_P} = x$ so we can replace x^{Δ_P} with x .
2. Therefore, if we have an equation where one side looks like x^{Δ_P} and we want to get x by itself, we can take the base both sides log of P .

3. $P = B^{B\Delta_P}$, so we can replace $B^{B\Delta_P}$ with P .
4. Therefore, if we have an equation where one side looks like $B\Delta_P$ and we want to get P by itself, we can raise B to the both sides.
5. $B = {}^{B\Delta_P}\sqrt{P}$, so we can replace ${}^{B\Delta_P}\sqrt{P}$ with B .
6. Therefore, if we have an equation where one side looks like $B\Delta_P$ and we want to get B by itself, we can take the both sides root of P .

3D: All of this has been kind of dumb

Let's list out the various equations we have solved in this chapter.

$$\begin{aligned}
 y^7 &= 128 \\
 7 &= \sqrt[3]{k} \\
 16^x &= 1,024 \\
 \sqrt[7]{78,125} &= 25 \\
 \log_2 x &= 9 \\
 \log_y 216 &= 3
 \end{aligned}$$

Now let's convert each of them to a ToP and resubject around the variable.

$$y^7 = 128 \rightarrow y \Delta^7 = 128 \rightarrow y \Delta_{128}^7 \rightarrow y = \Delta_{128}^7 \rightarrow y = \sqrt[7]{128}$$

$$7 = \sqrt[3]{k} \rightarrow 7 = \Delta_k^3 \rightarrow \Delta_k^3 \rightarrow \Delta_k^3 = k \rightarrow 7^3 = k$$

$$16^x = 1024 \rightarrow {}_{16}\Delta^x = 1024 \rightarrow {}_{16}\Delta_{1024}^x \rightarrow x = {}_{16}\Delta_{1024} \rightarrow x = \log_{16} 1024$$

$$\sqrt[7]{78,125} = 25 \rightarrow \Delta_{78,125}^7 = 25 \rightarrow \Delta_{78,125}^7 \rightarrow w = {}_{25}\Delta_{78,125}^7 \rightarrow w = \log_{25} 78,125$$

$$\log_2 x = 9 \rightarrow {}_2\Delta_x^9 \rightarrow {}_2\Delta_x^9 \rightarrow {}_2\Delta^9 = x \rightarrow 2^9 = x$$

$$\log_y 216 = 3 \rightarrow y \Delta_{216}^3 \rightarrow y \Delta_{216}^3 \rightarrow \Delta_{216}^3 = y \rightarrow \sqrt[3]{216} = y$$

In each case, we are able to solve the equation, or at least get it to the point where we can use a calculator for that last step, without ever doing any "I'm going to" statements, and without having to remember all of those insights about inverse operations. Once you get used to going straight from a traditional equation to a ToP with all three corners filled it, it gets even faster.

$$y^7 = 128 \rightarrow y \Delta_{128}^7 \rightarrow y = \sqrt[7]{128}$$

$$7 = \sqrt[3]{K} \rightarrow \begin{matrix} 3 \\ 7 \triangle K \end{matrix} \rightarrow 7^3 = K$$

$$16^x = 1024 \rightarrow \begin{matrix} x \\ 16 \triangle 1024 \end{matrix} \rightarrow x = \log_{16} 1024$$

$$\sqrt[w]{78,125} = 25 \rightarrow \begin{matrix} w \\ 25 \triangle 78,125 \end{matrix} \rightarrow w = \log_{25} 78,125$$

$$\log_2 x = 9 \rightarrow \begin{matrix} 9 \\ 2 \triangle x \end{matrix} \rightarrow 2^9 = x$$

$$\log_y 216 = 3 \rightarrow \begin{matrix} 3 \\ y \triangle 216 \end{matrix} \rightarrow \sqrt[3]{216} = y$$

OK, so, those were examples of the ways to solve the six different situations involving Level 3 Operations. Let's organize that a bit better, put it into a single table, and give general solutions without specific numbers.

$x^y = z$	$\begin{matrix} y \\ x \triangle z \end{matrix}$	$x = \sqrt[y]{z}$
$y^x = z$	$\begin{matrix} x \\ y \triangle z \end{matrix}$	$x = \log_y(z)$
$\sqrt[x]{y} = z$	$\begin{matrix} x \\ z \triangle y \end{matrix}$	$x = \log_z(y)$
$\sqrt[y]{x} = z$	$\begin{matrix} y \\ z \triangle x \end{matrix}$	$x = z^y$
$\log_y(x) = z$	$\begin{matrix} z \\ y \triangle x \end{matrix}$	$x = y^z$
$\log_x(y) = z$	$\begin{matrix} z \\ x \triangle y \end{matrix}$	$x = \sqrt[z]{y}$

Honestly, that didn't feel super organized to me. I mean, the table didn't even have headings. Also, repeatedly using x , y , and z might be kind of confusing, right? Who made that thing? The whole idea of that table was to always have the thing you are solving for be x . If you are OK with trying to solve for something else, and with picking which variable you want to solve for, this next table might be more useful.

Equation	Triangle	Desired variable	Solution
$x^y = z$	$\begin{array}{c} y \\ x \triangle z \end{array}$	x	$\sqrt[y]{z} = x$
		y	$\log_x z = y$
$\sqrt[a]{b} = c$	$\begin{array}{c} a \\ c \triangle b \end{array}$	b	$c^a = b$
		a	$\log_c b = a$
$\log_a R = S$	$\begin{array}{c} S \\ Q \triangle R \end{array}$	Q	$\sqrt[S]{R} = Q$
		R	$Q^S = R$

Now, that table might feel complete, right? I mean, no matter which of the three Level 3 Operations you encounter, and no matter which element you are trying to isolate, that table shows you how. Plus, you don't even really have to remember the table; you just have to remember how to move equations from traditional form to ToP form and back again. However, we are really just beginning (don't you groan at me!). Each of those equations has only one operation in it. What if there are multiple operations? We've looked at nested ToPs, but those can get a bit unwieldy. Also, what if there are other kinds of operations in the equation, like Level 1 and 2 Operations? In the next chapter, we will explore some additional tools to help us use ToPs in those situations.

3E: On the other hand, we did learn a few things

Wait, before we go on, I want to jump back to those insights we had before. Here they are:

$$B = \sqrt[x]{B^x} \quad x = \log_B B^x \quad P = (\sqrt[x]{P})^x \quad x = \log_{\sqrt[x]{P}} P \quad P = B^{\log_B P} \quad B = {}^{\log_B P} \sqrt{P}$$

Two of these are going to come in super handy later, so let's take a closer look at them and their ToP forms. To make them easier to read, I am going to insert some parentheses.

$$x = \log_B B^x$$

$$x = \log_B (B^x)$$

$$x = {}_B \triangle B^x$$

$$P = B^{\log_B P}$$

$$P = B^{\log_B (P)}$$

$$P = {}_B \triangle \log_B (P)$$

In this chapter, we talked about how to take the expressions from the right side and replace them with the simple variable on the left side. However, some situations will arise in which it will actually be useful to go in the other direction, to take a simple variable and replace it with a weird thingy that means the same thing. Why would we ever want to replace a simple thing with a more complicated thing? Well, consider the following arithmetic problem: $\frac{9}{7} - 1$. That's unpleasant by itself. However, consider the fact that 1 is the same thing as $\frac{7}{7}$. We can use that knowledge to replace the 1 in the subtraction problem with $\frac{7}{7}$, and it turns into $\frac{9}{7} - \frac{7}{7}$. If you remember how to add and subtract fractions, that's pretty easy to simplify into $\frac{2}{7}$. So you see, sometimes it is helpful to replace something simple with a more complicated (but equivalent) thingy. We'll come back to that later.

Chapter 4: Advanced ToP notation

4C: Benching

First of all, I know you're wondering why the first section in chapter 4 is 4C. Like, where did 4A and 4B go, right? So, what had happened was, I wrote a whole lot of stuff about this really cool thing you can do with ToPs called "chaining", but then I realized that I never really came back to it later, so, like, why keep it, right? Maybe there'll be a sequel one day, and I'll tell you all about the secret mysteries of chaining and ToP mutation (which is a whole other chapter I cut out). For now, though, the only advanced ToP notation I am going to tell you about is Benching.

Remember back in the middle of chapter 2, just before we talked about nesting, I brought in the idea of equations and expressions with multiple pieces, all strung together with different operations? I did this whole bit where I was like "sometimes a thing is actually two things, or three things, or even more". Does that sound familiar? I mean, it was not the best sentence, so I get it if you've forgotten.

Anyway, I only kind of half resolved that whole thing. See, nesting is good for when there's more than one Level 3 Operation, but what if there are other operations involved, operations from Level 1 or Level 2? There are a few different approaches to that. One method is to take the traditional form and replace all of the Level 3 Operations with ToP notation. Here are a couple of examples.

Two handwritten examples showing the replacement of Level 3 operations with ToP notation. The left example shows $x^2 + 3x + 2 = 0$ being rewritten as $x \triangle^2 + 3x + 2 = 0$. The right example shows $4x^4 - 7x^3 = 3\sqrt{y} + \log_{10}(z)$ being rewritten as $4(\triangle_x^4) - 7(\triangle_x^3) = \triangle_y^3 + \triangle_z^{\log_{10}}$.

When there is a ToP being multiplied by something, I like to put it in parentheses; I think that makes it easier to read. Now, I don't really have a name for the thing I just did there. It doesn't really need a name, right? I mean, it's just the way people normally write math, but now with ToPs in there. On the other hand, when you have a lower level operation embedded *inside* a Level 3, the traditional way of writing stuff doesn't meld well with the ToP. Here's an example of a lower level operation embedded inside a Level 3.

$$\sqrt[5]{Q + R}$$

If you try to write that as a single ToP, you might find that it's hard to tell exactly what the power is. Is it Q ? Is it R ? Is it both? To help with this, ToPs come with a special kind of grouping symbol called benches.

A diagram showing the conversion of the expression $\sqrt[5]{Q+R}$ into ToP notation. A red arrow points from the radical form to a ToP $\triangle^5$ with a horizontal bench underneath it, and the expression $Q+R$ is written on the bench.

The bench sticks out from one corner of the ToP and holds everything that's supposed to go in that corner. In this case, the power (the radicand) was $Q + R$, so that whole thing goes on the bench. The bench might look slightly different depending on which corner of the ToP it is on.

A diagram showing a ToP $\triangle^8$ with a bench on its bottom-left corner. The expression $2x+3z$ is written on the bench. A red arrow points from the expression $(2x+3z)^8$ to this notation.

A diagram showing a ToP $\triangle$ with a bench on its top-right corner. The expression $2a+b$ is written on the bench. A red arrow points from the expression 15^{2a+b} to this notation.

Of course, you can also do this with an entire equation, even when all three parts need benches.

A diagram showing an entire equation with ToPs and benches. The equation $(4a-b)^{xy+2} = 8z+5$ is shown on the left. A red arrow points to the right, where the equation is rewritten using ToPs: $(\triangle_{4a-b})^{\triangle_{xy+2}} = \triangle_{8z+5}$. Each part of the equation has its own bench.

This was a really short chapter, huh? Whatever. “Always leave them wanting more”, as the saying goes. Anyway, with this new tool of benching, we can take the so-called “laws of exponents” and do all sorts of fun stuff. That will be the topic of the next chapter.

Chapter 5: Converting Our Knowledge of Exponents

5A: The Laws of Exponents

From here, we are going to proceed on the assumption that you have learned the standard laws of exponents in their traditional form. The big idea here is to take those rules you (hopefully) already know and show what they look like in ToP notation. After that, we will use our skills at nesting, collapsing, recentering, resubjecting, and benching to show that there are a bunch of other rules you already know, but don't think you know because they look a little different. Okay, here's a standard list of exponent rules:

$$a^m \times a^n = a^{m+n}$$

$$(ab)^m = a^m b^m$$

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

$$a^m \div a^n = a^{m-n}$$

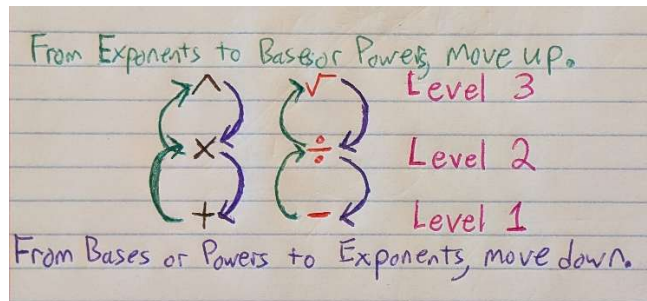
$$a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

$$a^{-m} = \frac{1}{a^m}$$

$$(a^m)^n = a^{mn}$$

$$a^0 = 1$$

There's a whole bunch of them there, I know. We're going to take a look at them one at a time. However, before that, there's this little visualization I want to share with you.



The idea here is that, whenever something is happening to an exponent, a base, or a power, you can mutate it so that something is happening to one of the other things, instead.

If you start with an operation on a base or a power and switch to the other one, the operation stays the same.

If you start with an operation on a base or a power and change it so there's an operation on the exponent, the new operation goes down a level.

If you start with an operation on an exponent and mess around to make something be happening to the base or power, the operation goes up a level.

Now, this diagram is missing fromation, gozintation, and logarithm, but I'm trying to give you a simple version to start. Actually, it's also missing a mysterious operation called **harmonization**, but we'll get to that later. Maybe.

5B: How to use that diagram

Let's consider this expression:

$$a^m \times a^n$$

What we have here is two power expressions, a^m and a^n , and they are being multiplied. Note that, in both of them, the base is a . The operation happening between the two powers is multiplication. Let's say we want to change it so the operation is happening to the exponents instead. Since we are going from operating on bases to operating on exponents, the level of the operation is going to decrease by one. Multiplication is a Level 2 Operation, so our new expression will have a Level 1 operation. Here's what that looks like:

$$a^{m+n}$$

This transformation is usually written as an equation.

$$a^m \times a^n = a^{m+n}$$

An important thing to remember here is that the two sides are equal. I mean, that's what an equation is, right? So, any time you see $a^m \times a^n$, you can replace it with a^{m+n} . Also, any time you see a^{m+n} , you can replace it with $a^m \times a^n$. Replacing multiplication of powers with addition of exponents corresponds to a purple arrow in the diagram. Going in the other direction, from *addition of exponents* to *multiplication of powers*, is represented by a green arrow in the diagram.

5C: Putting that first law in ToP form

Let's take that equation and put it in ToP form. There are three Level 3 Operations in it, and we're going to give each one its own ToP, at least to start. The one on the right side of the equals sign has a complicated exponent, so we'll have to use a bench there.

Look at the ToP form. On the right side, the addition is inside the exponent corner. On the left side, those two ToPs represent power expressions (because they are missing their power corners), so the left side has multiplication of powers.

You know what? Let's throw in some numbers so this looks a little more friendly. Let's let $a = 2$, $m = 3$, and $n = 4$. And before we even get into that part, let's look at 2^3 and 2^4 and play around with them a little bit.

Did you follow all that? The second to last step in each is really what you need. We discussed this at the end of chapter 3, kind of. That's where we discovered this rule:

OK, so, let's use 2, 3, and 4 to write out an example of the first law of exponents.

$$2^3 \times 2^4 = 2^{3+4}$$

Let's put the right side in ToP form, and then do just a little bit of simplification on the left side.

$$2^3 \times 2^4 = 2^{\overline{3+4}}$$

$$8 \times 16 = 2^{\overline{3+4}}$$

Now let's wrap it up in a ToP equation. This is going to require more benching.

$$\overline{3+4} \triangle 8 \times 16$$

Now let's resubject around the $3 + 4$.

$$3+4 = 2 \triangle 8 \times 16$$

Now we're going to use that rule from the end of chapter 3, the one where we take a simple thing and replace it with something more complicated (but still equivalent).

$$2 \triangle 2^3 + 2 \triangle 2^4 = 2 \triangle 8 \times 16$$

And finally, let's simplify a couple of little things on the left side.

$$2 \triangle 8 + 2 \triangle 16 = 2 \triangle 8 \times 16$$

The cool thing here is that, on the left side, there's addition, but on the right side, there's multiplication. This works with that diagram from the beginning of this chapter. On the right side, the multiplication is in the power, right? And on the left side, the two things that are being added are exponents. Think about that for a second. Remember, a whole ToP expression can be thought of as representing the thing in the missing corner. On the left side of the equation, both ToPs are missing their exponent corners, so each of them (each ToP) represents an exponent. Now, numbers are delightful, but let's write that out using symbols so we can be more general.

$$a \triangle x + a \triangle y = a \triangle xy$$

So, adding the exponents is the same as multiplying the powers. Wait, wasn't that just the first law of exponents? Jump back to the beginning of the chapter for a sec and verify. We started with this thing:

$$a^m \times a^n = a^{m+n}$$

$$a^{\triangle m} \times a^{\triangle n} = a^{\triangle m+n}$$

Look at that for a second. On the left side, those are two power expressions, right? And they are being multiplied? So that's multiplication of powers. And on the right side, there's addition going on inside the exponent, right? So that's addition of exponents. Now here's the part where I tell you that, in most math classes in most schools, those two equations are taught as *completely different ideas*. I guess we shouldn't really be too surprised, though. Math classes also tend to teach that addition and subtraction are two totally different things, but we know how intimately related they are. They're basically siblings, and not like the kind of siblings who dislike each other and only get together for holidays. I mean, not that there's anything wrong with that; it's just not what addition and subtraction are like. They have weekly video calls and share their thoughts and feelings. I really hope they also invite fromation; I wouldn't want anyone to feel left out. Wait, what was I talking about?

5D: Division and Subtraction

For my next trick, I will need the help of a few numbers from the audience. Ah, I see that Three and Five have volunteered! Thank you for your willingness to math. Anyway, check out these ToP equations:

Here, we have two trios of numbers. In both cases, the base is 3. On the left, the exponent is 5 and the power is 243. On the right, the exponent is 3 and the power is 27. Here are a couple more ways to express those same two arithmetic facts:

Take a moment to satisfy yourself that those facts are true, and then let's look at that second law of exponents from the beginning of the chapter.

$$a^m \div a^n = a^{m-n}$$

Applying that rule to our volunteer arithmetic facts, we can create the following equation.

Nice. Let's see what it looks like in ToP form, and then resubject.

Now, look back up at those various ways of expressing our volunteers. I'm going to replace several of the things in the equation with other things that are equal to them.

That was fun. OK, let's do it with symbols, now. This part's going to be a little more complicated, and involve some extra letters. Let's take the law $a^m \div a^n = a^{m-n}$ for starters. I am going to bring in x and y , because I want to. Let $x = a^m$ and let $y = a^n$. Here's that info in ToP equation form:

$$a^m \triangle x$$

$$a^n \triangle y$$

Now, we can take that law of exponents and put it in ToP form like this:

$$a^m \div a^n = a^{m-n} \rightarrow a^{m-n} \triangle a^m \div a^n$$

Now let's do some substitution again, using x and y .

$$a^{m-n} \triangle a^m \div a^n \rightarrow a^{a^m \triangle x - a^n \triangle y} \triangle x \div y$$

Now there is subtraction going on inside the exponent, right? And division inside the power. Wait, let's break it out into a regular equation (but still with ToPs).

$$a^{a^m \triangle x - a^n \triangle y} \triangle x \div y = a^{a^m \triangle x - a^n \triangle y}$$

Maybe it was easier to look at the other way around. Let's switch the right and left sides.

$$a^{a^m \triangle x - a^n \triangle y} = a^{a^m \triangle x - a^n \triangle y} \triangle x \div y$$

So, what do we have here? Subtraction of exponents corresponds to division of powers. Just like before, when we were looking at addition and multiplication, the Level 1 Operation on exponents corresponds to the Level 2 Operation on powers.

5E: Raising and Multiplication

So far, we've looked at moving between Level 1 and Level 2 Operations by changing whether we were operating on the exponent or the power. Next, let's look at moving between a Level 2 and a Level 3. Consider the two ToP equations below.

$$5^3 \triangle 25 = 5^3 \triangle 125$$

Now also take a look at these thingies, which are just resubjected versions of the one on the right. We're going to use these for some substitution in just a minute.

$$3 = 5 \triangle 125$$

$$125 = 5 \triangle 3$$

Now, the third law in the list at the beginning of the chapter says that $(a^m)^n = a^{mn}$. In words, if you have a power like a^m and you raise it to some value n , you can get the same thing by multiplying m and n to get a new exponent to stick on the original base. Nope, that was confusing; let me try again. Raising a power to a number is the same as multiplying its exponent by that number. Was that any better? No? Well, we were going to use numbers anyway, so let's throw in the 5, 3, and 2 for the a , m , and n , respectively.

$$(5^3)^2 = 5^{3 \cdot 2}$$

Let's start with one little substitution.

$$(125)^2 = 5^{3 \cdot 2}$$

Now let's switch over to ToP form and then resubject.

$$(125)^2 = 5 \overset{3 \cdot 2}{\Delta} \rightarrow 5 \overset{3 \cdot 2}{\Delta}_{125^2} \rightarrow 3 \cdot 2 = 5 \Delta_{125^2}$$

Now let's do the other substitution.

$$3 \cdot 2 = 5 \Delta_{125^2} \rightarrow (5 \Delta_{125}) \cdot 2 = 5 \Delta_{125^2}$$

Do you see the thing? On the left side, an exponent (the whole ToP) is being multiplied by 2. On the right side, the power (in the lower right corner) is getting raised to the 2. Cool. Now let's do it all again, but with symbols instead of numbers. First, though, I am going to need an extra letter.

$$\text{let } a^m = x$$

$$a \overset{m}{\Delta} x$$

$$m = a \Delta x$$

Now, let's take a look at that law of exponents.

$$(a^m)^n = a^{mn}$$

Let's do a substitution using our newly defined x .

$$x^n = a^{mn}$$

Unleash the triangle of power!

$$a \overset{mn}{\Delta} = x^n \rightarrow a \overset{mn}{\Delta} x^n \rightarrow a \overset{n \cdot m}{\Delta} x^n \rightarrow a \overset{n \cdot (a \Delta x)}{\Delta} x^n \rightarrow n(a \Delta x) = a \Delta x^n$$

Multiplying the exponent by a number is the same as raising the power to that same number. Is that what I said before? Honestly, I don't even remember.

5F: The Laws of Exponents Redux

I listed the laws of exponents back in section 5A, but I showed them the way they are traditionally written, and that's not really the point. Let's look at them in ToP form.

$$a^{\Delta^m} \cdot a^{\Delta^n} = a^{\Delta^{m+n}}$$

$$\underline{ab}^{\Delta^m} = a^{\Delta^m} \cdot b^{\Delta^m}$$

$$\underline{a \div b}^{\Delta^m} = a^{\Delta^m} \div b^{\Delta^m}$$

$$a^{\Delta^m} \div a^{\Delta^n} = a^{\Delta^{m-n}}$$

$$a^{\frac{m \div n}{\Delta}} = \Delta_n^m a^{\Delta^m}$$

$$a^{-\Delta^m} = 1 \div a^{\Delta^m}$$

$$a^{\Delta^m \Delta^n} = a^{\Delta^{mn}}$$

$$a^{\Delta^0} = 1$$

We discussed the three laws shown in the left column, and derived the corresponding laws of logarithms. Did you notice that? Here, let me show you.

$$a^{\Delta_m} + a^{\Delta_n} = a^{\Delta_{mn}}$$

$$a^{\Delta_m} - a^{\Delta_n} = a^{\Delta_{m \div n}}$$

$$m(a^{\Delta_n}) = a^{\Delta_m \Delta_n}$$

Chapter X: Farewell

Now, there are a lot more fun little identities you can find, more tools you can employ to manipulate ToP expressions. However, that is beyond the scope of this text. In fact, I've already gone far beyond the page count I had planned, and yet covered only a fraction of the content in my outline. Oh well. I guess my math was bigger than my stomach. What? Nevermind. Anyway, if you need me, I will be outside, shaking my fist at the darkness and crying out my regret that I was not able to solve all the problems of mathematics education. Adios.